Smart Energy Management and Load Optimization for Commercial Microgrid Systems

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***Abstract*—The increasing penetration of renewable energy sources in commercial buildings has amplified the need for intelligent energy management systems capable of coordinating distributed generation, battery storage, and heterogeneous loads. Commercial microgrids integrating solar photovoltaic (PV) systems, battery storage, and grid supply frequently suffer from inefficient energy utilization due to fragmented monitoring platforms, limited automation, and the absence of real-time decision-making. High-consumption loads such as Heating, Ventilation, and Air Conditioning (HVAC) systems and split-type air conditioners dominate energy usage in such environments, making their coordinated control critical to operational efficiency and cost reduction.**

This paper presents a comprehensive smart energy management and load optimization framework for commercial and residential microgrids. The proposed system enables real-time data acquisition from legacy, non-smart solar inverters using industry-standard Modbus RTU over RS485, and retrofits conventional HVAC and split AC systems for centralized monitoring and control through purpose-built hardware. A custom Home Assistant integration provides a unified interface for visualization, rule-based automation, and user interaction. A locally deployed AI-assisted decision-making module based on Retrieval-Augmented Generation (RAG) analyzes real-time energy data and generates actionable load control recommendations executable from the dashboard with a single interaction.

The system is validated through hardware deployment in a real commercial building environment. Experimental results demonstrate improved renewable energy utilization, reduced grid dependency, and responsive load prioritization, while maintaining low computational complexity and practical deployability. The framework offers a scalable, cost-effective pathway for small-to-medium commercial buildings to operate intelligently without replacing legacy infrastructure.

***Index Terms*—commercial microgrids, energy management systems, load optimization, HVAC control, Home Assistant, Modbus RTU, retrieval-augmented generation, ESP32, infrared control, renewable energy, legacy retrofitting**

I. Introduction

The global energy sector is undergoing a fundamental transition driven by escalating energy demand, intensifying climate concerns, and the rapid proliferation of renewable energy technologies. Solar photovoltaic (PV) systems have emerged as a cornerstone of distributed energy generation owing to their modularity, declining installation costs, and broad geographic availability [1]. In commercial buildings, PV systems are increasingly deployed alongside battery storage and utility grid interconnection, forming hybrid microgrids that improve resilience, reduce operational costs, and lower carbon footprints. Managing the energy flows within these hybrid systems intelligently remains, however, a significant and largely unsolved practical challenge.

Commercial buildings exhibit highly variable and demand-intensive load profiles dominated by HVAC systems, split air conditioners, lighting, and other critical loads. A significant fraction of total commercial building energy consumption—often exceeding substantial portions in warm climates—is attributable to air conditioning and climate control [2]. These loads are routinely operated independently of real-time energy availability data, resulting in excessive grid imports during peak generation periods, unnecessary battery cycling, and inflated electricity costs. Coordinating these loads with solar generation and battery state in real time represents the central energy optimization opportunity in commercial microgrid deployments.

A further complicating factor is the fragmented nature of monitoring and control in existing commercial installations. Solar inverters, HVAC controllers, and smart switches are typically managed through separate vendor-specific applications, preventing holistic system-level optimization. Most legacy in-

verters lack native network connectivity and must be interfaced through serial communication protocols such as Modbus RTU, which require custom hardware [3]. HVAC systems employ proprietary wired buses that require reverse engineering for EMS integration, and split AC units are controlled only via IR remote signals requiring custom blaster hardware for network-based control [4]. Advanced EMS proposals in the literature typically assume IoT-native devices and standardized APIs that are rarely available in such environments [5].

This work addresses these real-world challenges by presenting a practical, hardware-validated smart EMS tailored for commercial hybrid microgrids. The framework is built on three design principles: retrofitability, making legacy devices network-controllable without replacement; real-time closed-loop control, enabling automated load decisions based on live energy state; and user-centricity, preserving operator agency and occupant comfort at all times. The key contributions are:

- A custom ESP32-based inverter data logger using RS485/Modbus RTU for real-time parameter acquisition from legacy non-smart inverters.
- Reverse-engineered HVAC wired controller hardware and compact infrared blaster modules for direct remote control of HVAC units and split AC systems from Home Assistant.
- A unified Home Assistant integration providing real-time dashboard monitoring, rule-based automation, and historical energy logging for the complete microgrid.
- A locally deployed RAG-based AI decision module generating concise system summaries and one-click executable load optimization recommendations without cloud dependencies.

The remainder of this paper is organized as follows: Section II reviews related work and identifies gaps the proposed system addresses. Section III describes the layered system architecture. Sections IV and V detail hardware and software implementation respectively. Section VI presents the AI decision-making framework. Section VII reports experimental evaluation results. Sections VIII and IX provide discussion and conclusions.

II. Background and Related Work

A. Commercial Microgrid Configurations

Microgrids are localized energy systems integrating DERs, controllable loads, and storage within a defined boundary, operating either grid-connected or islanded [6]. Solar PV-based microgrids fall into three categories. Off-grid systems rely entirely on batteries and backup generation for uninterrupted operation, offering independence at high capital cost. On-grid (grid-tied) systems export surplus PV energy to the utility but lose all generation capability during grid outages due to anti-islanding protection. Hybrid systems combine PV, batteries, and grid connectivity, providing the flexibility of both configurations and enabling feed-in tariff benefits while maintaining resilience during outages [7]. The commercial testbed in this work operates as a hybrid microgrid, representing the most common and operationally demanding configuration for medium-scale commercial deployments.

B. Communication Protocols for Inverter and Load Integration

Reliable communication between energy assets and management platforms is prerequisite to real-time EMS operation. For solar inverters, RS485 paired with Modbus RTU is the industry standard, offering robust multi-device communication over extended distances [3]. The SunSpec Alliance has standardized Modbus register maps to improve cross-vendor interoperability. RS232, retained by older inverter models, supports only point-to-point communication at shorter distances. Modbus TCP extends the protocol over Ethernet for large-scale network-based deployments. At the application layer, Modbus RTU encodes inverter parameters including PV power, grid import/export, battery SOC, and load consumption in 16-bit registers with CRC error checking, providing a reliable basis for EMS data acquisition [8].

Commercial HVAC systems employ proprietary wired buses such as Daikin P1/P2 and Mitsubishi M-Net, posing significant integration challenges due to closed specifications. Two-wire and four-wire wired controller interfaces offer a more accessible integration pathway through protocol reverse engineering. For split-type air conditioners, IR remote control with manufacturer-specific pulse-width-modulated command sets is the dominant interface, requiring custom IR blaster hardware for EMS integration [4]. This diversity of interfaces necessitates a multi-modal hardware approach in the proposed system.

C. Energy Management and Load Optimization: State of the Art

Energy management in microgrids has attracted substantial research across three broad paradigms: optimization-based dispatch, AI-driven forecasting, and demand-side management. Multi-objective optimization using adaptive genetic algorithms minimizes cost, emissions, and power losses in microgrid dispatch, handling discrete decision variables effectively [9]. Commercial simulation tools such as the ABB Energy Simulator provide planning-oriented platforms for ROI estimation but are not designed for online closed-loop control [10].

AI-based frameworks combining metaheuristic optimization with graph-based learning have achieved high load forecasting accuracy in smart building microgrid simulations [11]. However, their multi-stage pipeline demands computational resources that may challenge edge deployment scenarios. Bio-inspired algorithms including Bat Optimization, Salp Swarm, and Cat Swarm methods have demonstrated peak demand reduction through shiftable load scheduling [12], but do not address direct real-time control of commercial HVAC equipment.

Reinforcement learning approaches using deep Q-networks and actor-critic methods have demonstrated adaptive microgrid dispatch policy learning in simulation [13]. Model predictive control (MPC) frameworks incorporating weather and demand

forecasts enable anticipatory scheduling of heating and cooling loads [14]. Despite their sophistication, these methods typically require extensive training data and remain predominantly validated in simulation with idealized communication assumptions. Rule-based automation in platforms such as Home Assistant offers a lightweight, practically deployable alternative for threshold-driven load management [15].

Demand response in smart grids has been comprehensively surveyed, with load shifting, direct control, and price-responsive strategies identified as primary mechanisms for grid interaction [16]. Distributed EMS architectures that combine local rule-based control with centralized monitoring have shown practical deployment viability in residential and small commercial settings [17].

D. Limitations of Existing Solutions and Contribution of This Work

A consistent limitation across existing solutions is their simulation-centric validation, which may overlook practical challenges such as communication latency, sensor noise, and device firmware variability [9], [11], [13]. Many proposals assume IoT-native infrastructure and standardized APIs, making them directly incompatible with the legacy equipment prevalent in small-to-medium commercial buildings [5], [12]. Computational complexity of state-of-the-art models can preclude edge deployment [11], [14], and user interaction and manual override mechanisms are frequently absent, potentially reducing real-world acceptability.

The proposed system directly addresses these gaps through hardware-validated deployment in a real commercial building. It retrofits non-smart inverters via RS485/Modbus RTU, enables direct HVAC and split AC control through custom hardware, provides a unified operator interface via Home Assistant, and delivers AI-assisted recommendations through a locally hosted RAG module—all without cloud dependencies or IoT-native equipment.

III. System Overview

The proposed smart EMS is organized into three functional layers that enforce a clean separation of concerns: data acquisition, decision-making, and load control. This architecture enables modular development, independent component validation, and scalability to accommodate additional devices and building sections without structural redesign of existing layers.

A. Data Acquisition Layer

This layer collects real-time energy parameters from the solar microgrid. Legacy solar inverters are interfaced via RS232 or RS485 with Modbus RTU. A custom ESP32-based hardware node acts as a data logger, polling inverter registers at regular intervals to extract PV generation power, grid import/export per phase, battery SOC and current, load consumption, AC phase voltages, and cumulative energy metrics. Data is transmitted wirelessly over Wi-Fi to the Home Assistant platform for visualization, automation, and AI query processing. A supplementary power monitoring node at the main distribution board provides per-phase load data for non-inverter-supervised circuits.

B. Decision-Making Layer

This layer processes real-time data through two complementary mechanisms. A rule-based automation engine in Home Assistant evaluates configurable threshold conditions—such as minimum battery SOC, surplus PV availability, or time-of-use pricing windows—and triggers predefined load control actions autonomously. The AI-assisted RAG module provides higher-level contextual analysis, generating natural language recommendations for operator review. Together, these mechanisms address both reactive automation for time-critical conditions and advisory intelligence for operator-guided optimization.

C. Load Control Layer

The load control layer interfaces with controllable loads through custom hardware. HVAC systems are controlled via ESP32-based wired controller units interfacing with the native HVAC bus. Split AC units are controlled via IR blaster modules. Smart switches with native Home Assistant support handle remaining loads. All devices are represented as Home Assistant entities, enabling unified automation, scheduling, and manual override from a single interface accessible from desktop and mobile.

IV. Hardware Implementation

A. Inverter Data Acquisition Hardware

The inverter data acquisition module is built around the ESP32 microcontroller, selected for its dual-core processing, integrated Wi-Fi, and extensive peripheral support. The hardware includes an RS232-to-UART converter for DB9-interface inverters and an RS485-to-UART converter for RJ45-bus inverters, both interfacing to the ESP32 UART port via a hardware multiplexer enabling automatic interface selection. The Modbus RTU master firmware polls a configurable register set at regular intervals, extracting PV input parameters, grid phase measurements, battery state, and energy totals. Parsed values are published to the Home Assistant MQTT broker over Wi-Fi. The module supports automatic entity discovery in Home Assistant and OTA firmware updates, eliminating the need for physical access during maintenance.

B. HVAC Control Hardware

Commercial HVAC wired controllers connecting indoor and outdoor units via proprietary 2-wire or 4-wire buses were reverse-engineered through logic analyzer protocol capture. The decoded command structure enabled design of a custom ESP32-based controller that replicates the native wired interface, operating in parallel with the original controller as a fallback without displacement. A capacitive touch panel enables local manual operation with real-time display of setpoint, operating mode, ambient temperature, and humidity. HVAC states and control commands are exposed as Home Assistant climate entities, enabling full remote control, scheduling, and automation. The hardware is housed in a standard electrical

enclosure suitable for installation within the existing HVAC control cabinet.

Fig. 1. Custom ESP32-based HVAC controller hardware with touch interface for manual operation and wireless Home Assistant integration.

C. Split AC Control Hardware

Split-type air conditioners present a distinct integration challenge due to the exclusive use of IR remote control with no exposed wired bus. The solution employs compact IR blaster modules, each comprising an ESP32 and multiple high-power IR transmitters arranged for omnidirectional coverage. A learned IR code library stores complete command frames for each supported AC brand and model, enabling control of power state, operating mode, temperature setpoint, fan speed, and swing settings. Multiple transmitters on each module enable simultaneous control of several units within a room. Modules are powered from standard USB supplies and appear in Home Assistant as climate entities, effectively converting IR-only legacy AC units into fully integrated smart loads without any hardware modification to the AC units themselves.

Fig. 2. Infrared blaster module for split AC control. Multiple transmitters allow simultaneous management of several units in a single room.

D. Power Monitoring Node

A dedicated power monitoring node deployed at the main distribution board extends energy visibility beyond the inverter. The node employs current transformer sensors on each phase line and an energy metering AFE IC interfaced to an ESP32 for calculation of per-phase active power, reactive power, power factor, and cumulative consumption. This data complements inverter metrics for complete building energy visibility and supports demand analysis and load disaggregation tasks that inform both rule-based automation thresholds and AI recommendation generation.

V. Software Architecture

A. Firmware for Smart Controllers

All custom devices run firmware that exposes them as discoverable entities over Wi-Fi, enabling seamless monitoring and control through Home Assistant.

B. Home Assistant Integration

A custom Home Assistant integration consolidates inverter parameters, HVAC and AC states, power monitoring data, and load control commands into a unified operational environment. All inverter register values are exposed as sensor entities with appropriate units, device classes, and state classes, enabling automatic energy dashboard population and long-term statistics storage. Custom service calls allow the automation engine and AI module to issue load control commands programmatically.

The operational dashboard presents a real-time overview of PV generation, grid import/export, battery SOC, and aggregate load power alongside individual HVAC and AC control panels. Lovelace UI cards render energy flow diagrams, time-series consumption charts, and direct control buttons in an intuitive layout accessible from browser and mobile app. The automation engine implements rule-based load control policies such as reducing HVAC setpoints when battery SOC falls below configurable thresholds, or disabling non-essential loads during grid outage conditions.

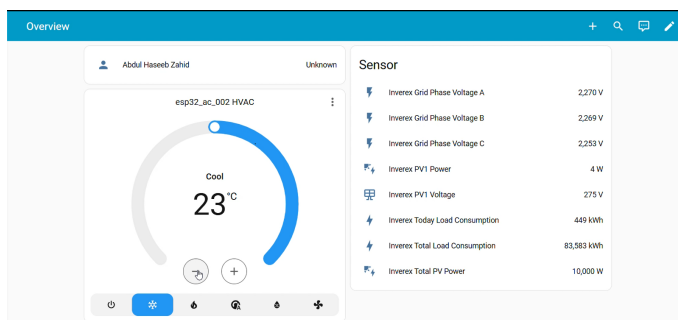

Fig. 3. The Home Assistant interface displaying real-time data from the inverter, along with HVAC and AC control interfaces, demonstrating the unified monitoring and control in the implemented system.

VI. AI-Assisted Decision Making

A. Retrieval-Augmented Generation Framework

The AI decision-making module employs a Retrieval-Augmented Generation architecture to produce contextually grounded, actionable energy management recommendations. RAG was selected over fine-tuned model approaches because it separates domain knowledge from language model parameters,

enabling the knowledge base to be updated as operational rules and tariff structures evolve, without model retraining—a critical operational advantage.

A proprietary rule database encoding expert knowledge about microgrid operation—including load priority matrices, battery protection thresholds, peak-shaving strategies, and comfort-preservation constraints—is converted into dense vector embeddings using a locally deployed sentence transformer model. These embeddings are indexed in a vector database hosted on the same server as Home Assistant. At each query cycle, current real-time parameters from the inverter and load sensors are serialized into a structured natural language query string and used to perform semantic similarity search, retrieving the most contextually relevant operational rules and strategies.

B. Local Large Language Model Integration

The retrieved context, together with the real-time data query, is passed as a structured prompt to a locally hosted LLM running via a local inference server. The LLM is instructed to produce two outputs: a concise plain-language system status summary describing the current energy state and any anomalies; and a prioritized list of recommended load control actions, each annotated with an expected energy impact qualifier. Prompt engineering constrains the output to parseable structured format, preventing hallucinated actions not grounded in the retrieved context.

Recommendations are presented within the Home Assistant dashboard through a custom card displaying the system summary and ranked action list. Each recommended action is accompanied by a one-click execution button calling the corresponding Home Assistant service, enabling the operator to implement AI-guided optimizations without manual device navigation. This design preserves operator agency while substantially reducing the cognitive load of real-time energy management across a multi-device commercial installation.

The entire RAG pipeline—embedding inference, vector search, and LLM generation—executes locally on the Home Assistant server with no cloud API dependency. This guarantees data privacy, eliminates recurring service costs, and ensures operational continuity during internet outages—a critical reliability requirement for commercial building management systems.

VII. Experimental Evaluation

A. Deployment Environment

The complete system was deployed in a medium-scale commercial office building equipped with a hybrid solar microgrid comprising roof-mounted PV arrays with substantial combined peak capacity, a battery bank sized for typical commercial backup requirements, and a standard utility grid connection. The building houses multiple centralized HVAC units and split AC units distributed across individual offices and meeting rooms. The deployment operated continuously over an extended evaluation period covering peak solar generation days, overcast conditions, grid outage events, and both high-occupancy and low-occupancy periods.

B. Communication Reliability and System Responsiveness

The inverter data acquisition system achieved reliable Modbus RTU register read performance throughout the evaluation period, with occasional communication transients during inverter internal fault recovery being automatically resolved within subsequent polling cycles. Wi-Fi transmission from ESP32 nodes to Home Assistant exhibited consistently low latency, providing effectively real-time data visibility for automation and AI processing requirements.

HVAC wired controller units demonstrated responsive end-to-end command execution—from Home Assistant service call issuance to confirmed state change on the HVAC unit—with delays well within practical operational bounds. IR blaster-based split AC control achieved reliable execution for all tested brands and models in the deployment. Multi-unit simultaneous command execution enabled synchronized setpoint changes across multiple AC units in a single automation action, confirming the practical utility of parallel IR transmission hardware.

C. Energy Utilization Outcomes

Automated load prioritization demonstrably reduced grid import during peak solar generation windows. During high-PV-output periods with moderate load demand, HVAC setpoints were automatically adjusted to pre-cool spaces while surplus solar power was available, reducing subsequent peak afternoon grid draw. Battery utilization improved as non-essential loads were deferred to coincide with battery discharge cycles rather than drawing simultaneously from grid and battery sources. The power monitoring node provided granular demand visibility that enabled more accurate automation threshold calibration than inverter-level aggregate data alone.

Qualitative operator feedback confirmed that AI-generated recommendations were consistently actionable and aligned with expected operational best practices. Operators routinely executed the top recommended action without modification, indicating strong relevance of the RAG-retrieved context to the prevailing energy state. The locally hosted LLM generated responses with latencies acceptable for the advisory use case where immediate sub-second response is not required.

VIII. Discussion

The proposed framework demonstrates that effective commercial microgrid optimization does not necessarily require computationally intensive forecasting models or cloud-dependent AI services. By focusing on direct control of the dominant energy consumers—HVAC and split AC systems—the system achieves meaningful improvements in renewable utilization and grid dependency reduction with a lightweight, locally executable software stack. The RAG-based AI module provides a practical middle ground between fully manual operation and fully autonomous control, augmenting

operator decision-making while preserving accountability and override capability.

A key differentiator of this work relative to existing literature is its grounding in real hardware constraints. The protocol reverse engineering required for HVAC bus integration, the practical considerations of IR-based AC control, and the communication robustness requirements of RS485 deployments in commercial RF environments are all addressed through validated engineering solutions rather than assumed away. This orientation toward deployability makes the framework immediately applicable to the installed base of commercial buildings without equipment replacement, significantly lowering the adoption barrier compared to IoT-native or cloud-dependent alternatives.

The modular architecture supports incremental expansion. Occupancy sensors, smart plugs, EV charging points, and sub-metering nodes can be integrated as Home Assistant entities and incorporated into the automation and AI recommendation framework without structural changes. The vector knowledge base can be updated with new operational rules as experience accumulates, improving recommendation quality continuously. The framework is also platform-agnostic at the inverter level—any inverter exposing a Modbus RTU register map can be onboarded by configuring the appropriate register addresses in the ESP32 firmware, without hardware changes.

Current considerations include the absence of predictive scheduling based on weather forecasts or occupancy patterns, which would enable proactive pre-conditioning and improved battery dispatch planning. The evaluation was conducted at a single site, and generalizability across building types, climatic zones, and inverter models would benefit from broader field study. Quantitative energy savings characterization over extended multi-month periods represents an important direction for future investigation.

IX. Conclusion and Future Work

This paper presented a deployable, hardware-validated smart energy management and load optimization framework for commercial and residential microgrids. The proposed system successfully retrofits legacy solar inverters for real-time data acquisition via Modbus RTU and extends network control to HVAC and split AC systems through custom wired and infrared hardware, all unified through a Home Assistant integration. The locally deployed RAG-based AI module augments operator decision-making with actionable, context-aware recommendations without cloud dependencies or high computational overhead.

Hardware deployment in a real commercial building confirmed reliable Modbus RTU communication, responsive load control command execution, and qualitatively validated AI recommendation relevance. The framework offers a practical, cost-effective pathway for small-to-medium commercial buildings to transition toward intelligent, renewable-optimized operation without replacing existing legacy infrastructure—addressing a critical gap between the theoretical sophistication of academic EMS proposals and the pragmatic constraints of real-world commercial deployment.

Future work will pursue three directions: integration of weather forecast and occupancy data into a predictive scheduling layer for proactive pre-conditioning and improved battery dispatch; expansion of the controllable load portfolio to include smart plugs, occupancy sensors, and EV charging infrastructure; and multi-site longitudinal field studies across diverse building types and climatic zones to further validate generalizability of the proposed approach.

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